

RATE & EXTENT – EQUILIBRIUM & LE CHATELIER

KEY VOCABULARY

Dynamic equilibrium: forward and backward rates are equal (closed system).

Le Chatelier: a system shifts to oppose any change made to it.

Position of equilibrium: whether more products or reactants are present.

Yield: the amount of product at equilibrium.

COMMON MISCONCEPTIONS

~~At equilibrium the reaction has stopped.~~

✓ Both reactions continue at equal rates.

~~Equilibrium means equal amounts.~~

✓ It means equal RATES, not equal amounts.

~~A catalyst shifts the equilibrium.~~

✓ A catalyst speeds up both sides equally — no shift.

LE CHATELIER'S PRINCIPLE

Increase concentration of a reactant → shifts right.

Increase pressure → shifts to the side with fewer gas moles.

Increase temperature → shifts in the endothermic direction.

The system acts to oppose the change.

A catalyst does NOT change the position.

1

DEFINE

State what is meant by **dynamic equilibrium**. [2 marks]

2

TEMPERATURE

The forward reaction is exothermic. **Predict** the effect of raising the temperature on yield. [2 marks]

3

PRESSURE

Predict how increasing pressure affects an equilibrium with fewer moles on the right. [2 marks]

4

CONCENTRATION

State what happens to the position if more reactant is added. [1 mark]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

At equilibrium the forward and backward ____ are equal, and the system shifts to ____ any change.

oppose

rates

amounts

support

6

CATALYST

Explain why a catalyst does not change the equilibrium yield. [2 marks]
